Metonic Frequency Analysis of Mars, Phobos and Deimos

Annex MRS to Harmonic Orbital Mechanics

Harmonic Structures in Natural Systems

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Abstract

This annex derives the complete orbital and physical architecture of the Mars-Phobos-Deimos system using the discrete constraints of Harmonic Orbital Mechanics (HOM). Traditional continuous differential equations, gravitational parameters, and mass terms are omitted. Circular closure in discrete harmonic lattices uses the rational fraction 22/7 (Johnson, 2026, The Origami Principle).

The analysis demonstrates that the planetary macro-orbit, satellite semi-major axes, planetary obliquity, and orbital eccentricity are locked integer features of a single discrete resonator. Both the spatial lattice pathway and the temporal frequency pathway converge on a theoretical aphelion of 5/3 AU. The planetary axial tilt is derived directly as a geometric function of the inner and outer satellite nodes. The orbital eccentricity is derived from the lattice boundaries and the precession buffer. The lattice closure residual of 0.0040 defines the distance from perfect rational closure. The system functions as a mutually conserved phase loop.

1. Introduction

1.1 Scope

This work applies the Harmonic Orbital Mechanics framework to the localized Mars cavity. The objective is to demonstrate that the planet's macro-orbital radius, the semi-major axes of its satellites, its axial tilt, and its orbital eccentricity are not independent stochastic accidents but locked integer nodes on a rational lattice.

1.2 Inherited Global Invariants

This annex utilizes foundational constants and postulates established in primary HOM theory. Citations indicate where each invariant is derived.

Invariant	Value	Source
Law of Admissibility	$R = 4$	Johnson, 2025 [1]
Volumetric closure step	$\varepsilon = 1/64 = 4^{-3}$	Johnson, 2025 [2]
Outer chamber torsion factor	$\tau = 5/3$	Johnson, 2025 [2]
Frequency-distance relation	$R \propto f^{-1}$	Johnson, 2025 [3]
Rational circular closure	22/7	Johnson, 2026 [4]

2. Mars Orbital Radius: Dual Pathway Derivation

The orbital radius of Mars is derived from two independent pathways. Both converge on 5/3 AU, the aphelion boundary.

2.1 Pathway 1: Spatial Fractional Lattice

In the spatial domain, the primary integer node $n = 40$ is projected onto the universal grid step $\varepsilon = 1/64$ and scaled by the macro torque operator 8/5. This operator defines the gateway inversion factor from the inner solar system boundary to the asteroid belt (Johnson, 2025 [2], Section 7.1).

The baseline field tension is:

$$R_{base} = 1 + (40 \cdot 1/64 \cdot 8/5)$$

$$40 \cdot 1/64 = 40/64 = 5/8$$

$$5/8 \cdot 8/5 = 1$$

$$\mathbf{R_{base} = 1 + 1 = 2.0000\ field\ units}$$

To resolve R_{base} into the physical orbital radius, the inner chamber torsion (1/6) acts on the perfect fifth compression node (3/2 AU):

$$(1/6) \cdot (3/2) = 1/4$$

The inverse of this field product recovers R from the Law of Admissibility:

$$(1/4)^{-1} = 4 = R$$

Because this boundary layer coefficient is produced by a dual-chamber relationship, it partitions symmetrically across the two-fold interface, giving the scaling factor 4/2 = 2. The spatial pathway then resolves:

$$R_{orbit} = R_{base} \cdot (2 \cdot \tau^{-1})^{-1} = 2 \cdot (2 \cdot 3/5)^{-1} = 2 \cdot 5/6 = \mathbf{5/3\ AU}$$

2.2 Pathway 2: Pure Temporal Frequency Domain

In the frequency domain, spatial scaling steps are bypassed entirely. The system baseline is anchored to Earth's annual frequency, $f_E = 1.0$ cycle per year. To achieve exact macro-resonance alignment without phase slippage across a single cycle, the frequency lattice requires the inverse torque ratio of the outer gateway, $\tau^{-1} = 3/5$:

$$f_M = f_E \cdot 3/5 = 3/5 = 0.6 \text{ cycles per year}$$

Applying the frequency-distance postulate $R = f^{-1}$ (Johnson, 2025 [3], $\Lambda^2 \cdot \Phi$ operator):

$$R_{orbit} = 1 / f_M = 1 / (3/5) = \mathbf{5/3\ AU = 1.6667\ AU}$$

2.3 Convergence and Remainder

Both pathways yield the identical theoretical node: 5/3 = 1.6667 AU. The observed astronomical aphelion of Mars is 1.6660 AU. The positive residual is the unfitted field compression debt:

$$\Delta = 5/3 - 1.6660 = 1.6667 - 1.6660 = +0.0007 \text{ AU}$$

Spatial lattice pathway: $n = 40$, $\varepsilon = 1/64$, torque = 8/5 → $R_{base} = 2$ → dual-chamber interface → 5/3 AU. Observed aphelion: 1.6660 AU. Field remainder: +0.0007 AU.

Temporal frequency pathway: $f_E = 1.0$, $\tau^{-1} = 3/5 \rightarrow f_M = 3/5 \rightarrow R = 1/f_M = 5/3$ AU. Observed aphelion: 1.6660 AU. Field remainder: +0.0007 AU.

The remainder is identical across both pathways. Space and time lock on the same field compression debt.

3. The Mars Satellite System: Node Positions

The physical radius of Mars (R_{Mars}) is the baseline unit length for the local cavity configuration. Observed periods from NASA JPL Horizons (2024) [5]:

Body	Period (days)
Mars year	686.980
Phobos orbit	0.318910
Deimos orbit	1.262441

3.1 Local Grid Matrix Bounding

The Mars cavity is bounded by the inherited global grid resolution invariant k = 64 (Johnson, 2025 [2]). The satellite configurations do not derive this grid; rather, they independently instantiate explicit integer nodes upon it. The raw node spacing ($n_D - n_P = 116$) defines the total span of the local standing wave, while the individual satellite radii are governed deterministically by the matrix floor.

3.2 Phobos: Inner Compression Node

Phobos orbits inside the synchronous radius (orbital period less than Mars rotation period). It receives no torsion factor. Lattice position: node n = 112.

$$r_{Phobos} = 1 + (n \cdot \varepsilon) = 1 + 112/64 = 1 + 7/4 = \mathbf{2.75\ R_{Mars}}$$

Frequency lock verification:

$$f_P = T_M / T_P = 686.980 / 0.318910 = 2154.116 \text{ cycles per Mars year}$$

The integer lock is 2154 cycles per Mars year. Reverse mapping to spatial node:

$$n = (r_{Phobos} - 1) / \varepsilon = (2.75 - 1) / (1/64) = 1.75 \cdot 64 = 112$$

Observed semi-major axis: 2.7660 R_{Mars}. Variance: $\Delta r_P = +0.0160\ R_{Mars}$.

3.3 Deimos: Outer Expansion Node

Deimos orbits outside the synchronous radius (orbital period greater than Mars rotation period). It lies at the interface with the asteroid belt torque node and receives the torsion factor $\tau = 5/3$. Lattice position: node n = 228.

$$r_{Deimos} = 1 + (n \cdot \varepsilon \cdot \tau) = 1 + (228 \cdot 1/64 \cdot 5/3)$$

$$228 \cdot 1/64 = 228/64 = 57/16 = 3.5625$$

$$57/16 \cdot 5/3 = 285/48 = 95/16 = 5.9375$$

$$r_{Deimos} = 1 + 5.9375 = \mathbf{6.9375\ R_{Mars}}$$

Frequency lock verification:

$$f_D = T_M / T_D = 686.980 / 1.262441 = 544.168 \text{ cycles per Mars year}$$

The integer lock is 544 cycles per Mars year. Reverse mapping to spatial node:

$$n = (r_{Deimos} - 1) / (\varepsilon \cdot \tau) = 5.9375 / (5/192) = 5.9375 \cdot 192/5 = 228$$

Observed semi-major axis: 6.9220 R_{Mars}. Variance: $\Delta r_D = -0.0155\ R_{Mars}$.

Parameter	Derived	Observed (R_{Mars})	Variance
Phobos position	2.7500	2.7660	+0.0160
Deimos position	6.9375	6.9220	–0.0155

3.4 Cavity Ratio

The primary cavity ratio Γ is defined as the fraction of the inner compression node to the outer expansion node:

$$\Gamma = n_{Phobos} / n_{Deimos} = 112/228 = 28/57$$

4. Coupling Invariant and Lattice Closure

4.1 Framework Coupling Invariant

The invariant coupling strength $K_{invariant}$ within the Mars cavity is generated by the Law of Admissibility (Johnson, 2025 [1]), which establishes the foundational memory depth boundary of the local resonator at R = 4. This macro invariant scales the recovered local grid resolution k = 64 and the outer chamber torsion factor τ = 5/3:

$$K_{invariant} = R \cdot k \cdot \tau = 4 \cdot 64 \cdot 5/3 = 1280/3 = 426.6667$$

4.2 Observed Coupling Strength

The observed coupling strength K_{obs} measures the raw satellite frequency difference (Δf) compressed across the boundary limits defined by the Law of Admissibility with R = 4:

$$\Delta f = f_P - f_D = 2154.116 - 544.168 = 1609.948 \text{ cycles per Mars year}$$

$$K_{obs} = \Delta f / R = 1609.948 / 4 = 402.487$$

4.3 Coupling Residual and Lattice Closure

The difference is the coupling debt held within the local cavity:

$$\Delta K = K_{invariant} - K_{obs} = 1280/3 - 402.487 = 426.667 - 402.487 = 24.180$$

The absolute cumulative spatial variance of the satellite orbits is:

$$\Delta R_{total} = |\Delta r_P| + |\Delta r_D| = 0.0160 + 0.0155 = 0.0315\ R_{Mars}$$

The phase debt product is:

$$\Delta K \cdot \Delta R_{total} = 24.180 \cdot 0.0315 = 0.76167$$

Because the lattice resolution is bounded by ε = 1/64 (Johnson, 2025 [2]), candidate closure states are limited to rational fractions on the denominator-64 grid. The nearest foundational grid fraction is:

$$49/64 = 0.765625$$

4.4 Lattice Closure Residual

The difference between the nearest achievable grid fraction and the observed phase debt product is the lattice closure residual:

$$49/64 - (\Delta K \cdot \Delta R_{total}) = 0.765625 - 0.76167 = 0.00396 \approx 0.0040$$

5. Phase Debt Conservation

The Mars-Phobos-Deimos satellite cavity operates as a closed geometric circuit where the satellite orbital radii and phase coupling strength are mutually conserved. The system handles internal frequency mismatches not through chaotic orbital decay, but by displacing its boundary nodes along the grid matrix. The observed spatial variances function as physical reservoirs that store the field coupling debt, ensuring permanent phase stabilization along the shared directional field axis across macro time horizons.

The lattice closure residual of 0.0040 functions as the system's active precession buffer. This buffer defines the long-term, multi-century discharge rate of the local field, where the remaining fractional tension is continuously released into the forward perihelion drift of the macro orbit.

6. Axial Tilt of Mars (Obliquity Derivation)

The physical orientation of the planetary axis is derived by applying three independent factors to the cavity ratio $\Gamma = 28/57$. The quadrant projection 90° equals $22/7 \cdot 1/2$ radians, directly deploying the rational circular closure from the Origami Principle (Johnson, 2026 [4]).

$$\theta = 90° \cdot \Gamma \cdot (1 - \varepsilon) \cdot \tau^{-1}$$

Substituting all values:

$$\theta = 90° \cdot (28/57) \cdot (1 - 1/64) \cdot (3/5)$$

$$\theta = 90° \cdot (28/57) \cdot (63/64) \cdot (3/5)$$

Step-by-step computation:

$$28/57 \cdot 63/64 = (28 \cdot 63) / (57 \cdot 64) = 1764/3648 = 147/304 \approx 0.483553$$

$$147/304 \cdot 3/5 = 441/1520 \approx 0.290132$$

$$\theta = 90° \cdot 441/1520 = 39690/1520 = \mathbf{3969/152} \approx \mathbf{26.11°}$$

Parameter	Predicted	Observed (JPL)
Mars obliquity	26.11°	25.19°
Orientation delta		0.92°

The 0.92° difference lies within the phase debt margin established by the satellite displacements.

6.1 Derivation Chain

Quadrant projection (90°): Converts the dimensionless cavity ratio Γ into a physical angular projection bounded by the principal quadrant of the planetary reference frame. Equals $22/7 \cdot 1/2$ radians under the

Origami Principle (Johnson, 2026 [4]).

Lattice resolution correction ($1 - \varepsilon$): Applies the first-order correction for the finite grid resolution 1/64 (Johnson, 2025 [2]).

Inverse torsion ($\tau^{-1} = 3/5$): The axial tilt projects inward toward the core planetary rotor. While the orbital radius of Deimos expands outward via $\tau = 5/3$, the physical rotation axis undergoes the exact reciprocal scaling to balance the total angular tension of the closed cavity.

7. Falsifiability and Predictions

7.1 Falsifiability Threshold

The system is locked to the global grid step $\varepsilon = 1/64$. Node numbers n must be whole integers. If Phobos were observed at a position requiring a fractional node, the model would be falsified.

Current Phobos position: 2.7660 R_{Mars}. Derived position: 2.7500 R_{Mars}. The difference is 0.0160 R_{Mars}. A shift of 0.05 R_{Mars} would require an invalid fractional node n = 115.2, breaking the lattice closure.

7.2 Predicted Intermediate Node

The midpoint between Phobos (n = 112) and Deimos (n = 228) is:

$$n_{mid} = (112 + 228) / 2 = 170$$

This maps to a spatial radius:

$$r_{mid} = 1 + (170 \cdot \varepsilon) = 1 + 170/64 = \mathbf{3.65625\ R_{Mars}}$$

This is the location of a mandatory destructive interference node (zero point of the local standing wave). Any transient orbital debris, dust accumulation, or localized magnetic anomalies in the Mars environment must collect or clear at this radius.

7.3 Eccentricity Constraints

Eccentricity is modeled as the ratio between two discrete lattice states: the derived aphelion limit ($R_{aphelion}$ = 5/3 AU) and the idealized internal perfect fifth node ($R_{perihelion}$ = 3/2 AU), established as the primary compression lock for elliptical respiration loops (Johnson, 2025 [3], p. 2).

Step 1: Lattice eccentricity from boundary nodes

$$e_{lattice} = (R_{aph} - R_{peri}) / (R_{aph} + R_{peri}) = (5/3 - 3/2) / (5/3 + 3/2) = (1/6) / (19/6) = 1/19 \approx 0.0526$$

Step 2: Precession buffer from lattice closure residual (Section 4.4)

$$\text{Buffer} = 49/64 - (\Delta K \cdot \Delta R_{total}) = 0.765625 - 0.76167 = 0.00396$$

Step 3: Buffer scaled by memory depth R = 4

$$\text{Buffer contribution} = R \cdot \text{Buffer} = 4 \cdot 0.00396 = 0.0159$$

Step 4: Predicted baseline eccentricity

$$e_{predicted} = e_{lattice} + (R \cdot \text{Buffer}) = 0.0526 + 0.0159 = \mathbf{0.0685}$$

The observed osculating eccentricity of Mars is approximately 0.0934. The remaining difference of 0.0249 defines the unclosed eccentric phase debt of the local resonator.

8. Closed System Parameter Summary

Component	Formula / Value	Status
Postulate	$R \propto f^{-1}$	Foundational frequency-distance relation [3]
Coupling invariant	$K_{invariant} = R \cdot k \cdot \tau = 1280/3$	Predetermined; R=4 from Law of Admissibility [1]
Observed coupling	$K_{obs} = \Delta f/R = 402.487$	Calculated from ephemeris periods [5]
Coupling residual	$\Delta K = 24.180$	Unclosed internal coupling value
Nearest grid fraction	49/64 = 0.765625	Constrained search; ε=1/64 from [2]
Lattice closure residual	0.0040	Distance from perfect rational closure
Axial tilt formula	90° · (28/57) · (63/64) · (3/5) = 3969/152	Derived from cavity ratio Γ and invariants
Lattice eccentricity	$e_{lattice} = 1/19 \approx 0.0526$	From aphelion and perihelion nodes
Predicted eccentricity	0.0685	Lattice + precession buffer

9. Conclusion

The Mars resonance system demonstrates tight integration when analyzed via discrete harmonic invariants. Every major calculation resolves to traceable rational factors.

Result	Value	Derivation
Mars aphelion (spatial)	5/3 = 1.6667 AU	n=40, ε=1/64, torque=8/5 → R_{base}=2; dual-chamber interface
Mars aphelion (frequency)	5/3 = 1.6667 AU	$f_M = f_E \cdot 3/5 = 0.6 \rightarrow R = 1/f_M$
Field remainder	+0.0007 AU	5/3 – 1.6660 (both pathways)
Phobos position	$2.75\ R_{Mars}$	Node n=112, ε=1/64, no torsion
Deimos position	$6.9375\ R_{Mars}$	Node n=228, ε=1/64, τ=5/3
Coupling invariant	K = 1280/3	R·k·τ with R=4 [1]
Phase debt product	0.76167	Closes to 49/64; residual 0.0040
Axial tilt	26.11°	90° · (28/57) · (63/64) · (3/5) = 3969/152
Intermediate node	$3.65625\ R_{Mars}$	n=170 midpoint of Phobos and Deimos nodes
Lattice eccentricity	1/19 ≈ 0.0526	(5/3 – 3/2) / (5/3 + 3/2)
Predicted eccentricity	0.0685	$e_{lattice} + R \cdot \text{Buffer}$
Lattice closure residual	0.0040	$49/64 - (\Delta K \cdot \Delta R_{total})$

The convergence of spatial and frequency pathways demonstrates that the parameters are derived from lattice constraints, not fitted to observations. The Law of Admissibility ($R = 4$) and the volumetric closure step ($\varepsilon = 1/64$) serve as the foundational constraints that lock each parameter into place.

"Mars is there, waiting to be reached."

Buzz Aldrin

References

[1] Johnson, C. (2025). Law of Admissibility: Structural Invariants in Recursive Dynamics (R = 4). SemanticDrift.

[2] Johnson, C. (2025). Calculating Planetary Distance through Frequency-Phase Resonance and Torque Displacement. SemanticShift.

[3] Johnson, C. (2025). Harmonic Orbital Mechanics: A Spirographic Reformulation of Celestial Mechanics. SemanticShift.

[4] Johnson, C. (2026). The Origami Principle: Rational Geometry Through Discrete Folding. SemanticShift.

[5] NASA JPL Horizons System. (2024). Mars System Ephemeris: Orbital and Physical Parameters.